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MID TERM EXAMINATION AUTUMN/SPRING, 2026

Branch & Academic Year: ECE, CSE 2024 Batch **Semester: IV**

Subject: Machine Learning **Subject Code: CS342E** **Instructor Code: AS**

Max Marks: 30 **Duration: 90 Minutes** **Date: 27/2/2026**

Section A-Theoretical Questions [20-Marks]. Attempt all questions. Answer to the point as per the marks allotted to each section.

Q1. (a) Explain the motivation behind evolution of Machine Learning modelling/programming with examples w.r.t. to traditional programming. [1.5]
Q1. (b) Define cost function of a linear regression model with one feature only and solve it to get the design parameters. Taking this as an example, explain why can't we have an analytical solution for any regression problem. [0.5+1+1]

Q2. Explain with examples the different assumptions under which linear regression modelling holds true. [2+2]

Q3. Explain the concept of regularization. Explain the regularization approach that can be used for feature selection along with regularization. Highlight its feature which makes it suitable for feature selection. [1+3]

Q4. Define overfitting and under fitting issues in Machine learning with examples. How they are related to Bias and Variance of a model? Explain the practical relevance of Bias-Variance trade-off with examples. Use suitable figures to explain properly. [1+2+1]

Q5. Following dataset is given to implement simple liner regression using gradient descent.
x= [1,2,4], y=[1,3,3]
Assumptions are:
- **Initial coefficient: zero**
- **Learning rate: 0.01**

Find out liner regression line after one epoch. [4]

Section B-MCQ [10-Marks]

Note: Each correct answer will be awarded one (1) mark and wrong (- 0.25).

1. If one item is fixed and unchangeable and the other item varies, the correlation coefficient will be:
(a) Positive (b) Negative (c) Zero (d) Undecided

2. In a simple linear regression model (One independent variable), If we change the input variable by 1 unit. How much output variable will change?
a) by 1 b) no change c) by intercept d) by its slope

3. Regression is a
1. **Numerosity reduction technique**
2. **Dimensionality reduction technique**
3. **Parametric way of data representation**

Options: A. 1 and 3 B. 1 and 2 C. 2 and 3 D. 1, 2 and 3